FREE, OPEN, FIELD-FIRST

Northern Hardwoods & Mixedwoods Dendrology and Silvics Intensive

Complete course syllabus

A rigorous, self-paced course in identifying and understanding the important woody trees of northern hardwood, Great Lakes-St. Lawrence, Acadian, boreal-transition, Allegheny, and associated mixedwood forests.

10 modules	**80 taxa**	**32 graded assessments**

Cochetopa.co/course/

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Release 2026.2

OVERVIEW

Course at a glance

This is a free, open-access, upper-division-style forestry course centered on durable field identification. There is no tuition, application, or password. Course materials may be browsed openly. Learners who want synchronized progress, grades, mastery records, and optional reminders enter an email address and confirm the link sent to that address.

Nominal length: 10 weeks. Actual pace: entirely self-directed. A learner may move intensively, work alongside a field season, or return in short sessions without losing the cumulative structure.

Geographic scope

Minnesota and the western Great Lakes; Wisconsin; Michigan; northern Illinois and Indiana where the northern mixedwood flora reaches them; Ontario through the Great Lakes-St. Lawrence region and north toward North Bay and Sudbury; southern Quebec; New York; Vermont; New Hampshire; Maine; New Brunswick; Massachusetts and Connecticut where relevant; Pennsylvania through the Allegheny Plateau; and northern and northwestern New Jersey.

The ecological center of gravity is northern hardwood forest, with deliberate preparation for boreal, conifer-hardwood, lowland, riparian, oak-hardwood, Allegheny, Acadian, and Great Lakes-St. Lawrence communities.

Who the course is for

Forestry students, field technicians, landowners, natural-resource professionals, arborists, advanced naturalists, and anyone who wants a demanding rather than casual approach to tree identification. No formal prerequisite is enforced, but the course assumes a willingness to learn morphology, scientific names, and applied ecology.

WHAT SUCCESSFUL LEARNERS CAN DO

Course objectives and learning outcomes

- Identify the 80 required taxa rapidly from unfamiliar photographs or field specimens when the evidence supports a species-level call.
- Use foliage, bark across age classes, buds and twigs, reproductive structures, whole-tree form, and ecological context as independent lines of evidence.
- Give the preferred common name or canonical scientific name for ordinary identification items and retrieve common-to-scientific and scientific-to-common names on dedicated nomenclature work.
- Name the nearest plausible alternative, point to a visible separating character, and request the most useful additional view when evidence is incomplete.
- Choose genus, group, or insufficient evidence rather than forcing a species identification that the specimen cannot defend.
- Apply concise field silvics: shade tolerance, site and moisture relationships, geographic affinity, regeneration, disturbance response, succession, browse, important health agents, and management implications.
- Recognize important age, disease, insect, injury, weather, lichen, browse, and growth-form variation without confusing host identification with condition diagnosis.
- Transfer recognition across northern hardwood, Acadian, Great Lakes-St. Lawrence, Allegheny, boreal-transition, lowland, riparian, and oak-transition contexts.
- Use repeated retrieval and an error ledger to correct persistent discrimination, nomenclature, silvics, and overconfidence errors.

The governing field standard

Identify the narrowest taxonomic rank the visible evidence can defend. Ecological context can strengthen or weaken a hypothesis, but it does not substitute for morphology.

Recognition from one familiar photograph is not mastery. Tier A and Tier B species recur across multiple organs and unfamiliar specimens before they can reach Multi-organ Mastery or Certified status.

INSTRUCTIONAL DESIGN

How the course teaches field recognition

Foliage	Arrangement, attachment, surfaces, margins, venation, lobing, petioles, fascicles, sun/shade and seasonal variation.
Bark	Sapling, intermediate, mature, old, wet, shaded, damaged, diseased, and lichen-covered bark.
Dormant material	Twig arrangement, buds, scales, scars, pith, pubescence, thorns, terminal buds, and persistent structures.
Reproductive	Samaras, acorns, nuts, drupes, pomes, capsules, cones, strobili, catkins, seeds, and attached or detached fruit.
Whole tree	Crown architecture, branching, bole form, regeneration habit, viewing distance, habitat, associates, and apparent site.
Condition	Host identity under disease, insects, injury, browsing, weathering, mechanical damage, and misleading secondary evidence.

Weekly learning cycle

- Session A - introduce a manageable set of species with diagnostic examples and ecological context.
- Session B - retrieve the new material without immediate answers.
- Session C - add taxa and compare realistic confusers directly.
- Session D - work across foliage, bark, reproductive, dormant, and whole-tree evidence.
- Session E - mix the week with earlier material and target weak species.
- Weekly practical - complete a graded, primarily unfamiliar-image assessment.

Image pools

Teaching, practice, and formal-examination images are kept conceptually separate. The public teaching and practice library contains more than 1,000 documented photographs. Formal assessments use separately held specimens so image memorization cannot substitute for species recognition.

COURSE CALENDAR

Weekly sequence: Weeks 1-5

Week	Instructional focus	Scheduled assessments
1	Foundations + Northern Hardwood Anchors Apply dendrological vocabulary and the maximum-defensible-rank decision rule.	Weekly practical (18 items, 35 min)
2	Northern Hardwood Associates Discriminate the hard-maple, understory-maple, paper-birch, dark-barked birch, cherry, and hornbeam sets.	Weekly practical (20 items, 40 min) Nomenclature check (20 items, 15 min) Silvics retrieval (12 items, 30 min)
3	Coniferous Mixedwoods Separate spruce, fir, hemlock, pine, larch, cedar, and juniper from needle attachment and arrangement.	Weekly practical (24 items, 48 min)
4	Boreal and Great Lakes Transition Identify boreal-transition Populus, alder, and mountain-ash taxa from organs and whole-tree form.	Cumulative practical (32 items, 64 min) Nomenclature check (30 items, 22 min) Silvics retrieval (16 items, 40 min) Field-context exercise (20 items, 45 min)
5	Ashes, Elms, Lowlands, and Riparian Systems Discriminate ashes and elms from foliage, buds, leaf scars, samaras, bark, and site.	Weekly practical (24 items, 48 min) Midterm examination (60 items, 120 min) Lowland field exercise (20 items, 45 min)

COURSE CALENDAR

Weekly sequence: Weeks 6-10

Week	Instructional focus	Scheduled assessments
6	Oaks and Oak-Northern Hardwood Transition Discriminate the regional red-oak and white-oak groups from leaves, buds, bark, acorns, and site.	Weekly practical (24 items, 48 min) Nomenclature check (40 items, 30 min) Silvics retrieval (20 items, 50 min)
7	Hickories, Walnuts, and Allegheny/Southern Associates Discriminate regional hickories and walnuts using leaf, bud, pith, nut, husk, and bark combinations.	Weekly practical (24 items, 48 min)
8	Acadian and Eastern Regional Integration Identify Atlantic white-cedar and pitch pine and place them against their closest regional confusers.	Weekly practical (28 items, 56 min) Nomenclature check (50 items, 38 min) Silvics retrieval (24 items, 60 min) Regional transfer exercise (24 items, 55 min)
9	Pathology, Variation, Winter Dendrology, and Adversarial Identification Discriminate black locust, tree-of-heaven, and white mulberry from native compound-leaved and variable-leaved trees.	Weekly practical (30 items, 65 min) Beech condition laboratory (24 items, 60 min) Winter/adversarial practical (30 items, 70 min)
10	Comprehensive Integration and Certification Demonstrate rapid, calibrated identification of all eligible taxa from unfamiliar single-organ and mixed evidence.	Scientific-name final (80 items, 60 min) Silvics synthesis (40 items, 100 min) Field-condition check (30 items, 70 min) Foliage final (20 items, 35 min) Bark final (20 items, 40 min) Reproductive final (16 items, 30 min) Dormant final (20 items, 45 min) Whole-tree/regional final (20 items, 40 min) Rapid mixed-evidence final (50 items, 70 min)

EVALUATION

Assessment, grading, and mastery

Category	Weight	Scope
Weekly identification practicals	**25%**	Nine cumulative practicals; the lowest receives reduced weight rather than disappearing.
Midterm examination	**15%**	Cumulative through Week 5.
Scientific nomenclature	**10%**	Common-to-scientific and scientific-to-common retrieval throughout the course.
Silvics	**15%**	Site, regeneration, succession, disturbance, condition, and management relationships.
Field-style and condition identification	**10%**	Bark variation, winter material, ecological context, health effects, and insufficient-evidence decisions.
Comprehensive final practical	**25%**	Primarily unfamiliar photographs and mixed evidence, including organ-specific batteries.

Identification response rule

For ordinary identification stations, one defensible identity is enough: either the preferred common name or the scientific name. Additional confidence and field-reasoning prompts may be used for calibration but are not required to submit. Dedicated nomenclature questions require the requested name direction.

Mastery states

New -> Learning -> Recognized -> Multi-organ Mastery -> Certified

A species cannot become Certified through repeated exposure to one organ. Mastery records track foliage, bark, reproductive, dormant, whole-tree, condition, nomenclature, silvics, confuser performance, confidence calibration, and spacing history.

Feedback and remediation

Meaningful mistakes enter an Error Ledger as knowledge, discrimination, nomenclature, overconfidence, or silvics errors. Weak and confidently wrong material returns more aggressively; strong material returns at wider intervals.

PRACTICAL DETAILS

Participation, technology, and course policies

Access and cost

The course is free. No payment information is requested. Lessons may be explored without creating a traditional account. To keep progress across devices, enter an email address and confirm the emailed link. There is no password. Optional encouragement emails remain under the learner's control.

Pacing

The ten-week sequence is instructional, not a deadline. Learners may move quickly or slowly. Cumulative retrieval remains active regardless of pace, and serious gaps trigger targeted remediation without preventing progress into later modules.

Technology

A current desktop or mobile browser and a reliable internet connection are recommended. A larger screen is useful for bark and whole-tree images. Progress saves on the current device; confirmed email access enables synchronization and the persistent gradebook. Printable weekly packets and a PDF syllabus support offline instruction and instructor-led use.

Assessment integrity

Formal assessment specimens and answer material should not be redistributed. Source attribution and licensing remain visible where appropriate, but private examination assets stay separate from the public teaching pool. The goal is honest field competence, not recognition of a memorized station order.

Scientific names and taxonomy

The syllabus lists the canonical course answer for every taxon. Important alternatives are documented in the course profiles. A documented accepted alternative is not penalized. Genus and specific epithet are required for full credit when a species-level scientific name is requested.

Condition and pathology

Health agents are included to strengthen dendrology rather than turn the course into forest pathology. American beech receives the most extensive condition sequence. Other recurring examples include emerald ash borer, hemlock woolly adelgid, elongate hemlock scale, butternut canker, Dutch elm disease, white pine blister rust and weevil, oak wilt, black knot, spruce budworm, balsam woolly adelgid, browse deformation, and fire or mechanical scars.

Instructor and course link

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https://cochetopa.co/course/

TAXONOMIC UNIVERSE

Master Species List - Tier A

Core mastery - dominant, widespread, ecologically important, or frequently encountered trees receiving the greatest repetition. 26 taxa.

Preferred common name	Canonical scientific name	Family	Week
American basswood	Tilia americana	Malvaceae; Tiliaceae in older texts	1
American beech	Fagus grandifolia	Fagaceae	1
American elm	Ulmus americana	Ulmaceae	5
American hornbeam	Carpinus caroliniana	Betulaceae	2
Balsam fir	Abies balsamea	Pinaceae	3
Bigtooth aspen	Populus grandidentata	Salicaceae	4
Black ash	Fraxinus nigra	Oleaceae	5
Black cherry	Prunus serotina	Rosaceae	1
Black spruce	Picea mariana	Pinaceae	3
Eastern hemlock	Tsuga canadensis	Pinaceae	1
Eastern hophornbeam	Ostrya virginiana	Betulaceae	1
Eastern white pine	Pinus strobus	Pinaceae	3
Green ash	Fraxinus pennsylvanica	Oleaceae	5
Northern red oak	Quercus rubra	Fagaceae	6
Northern white-cedar	Thuja occidentalis	Cupressaceae	3
Paper birch	Betula papyrifera	Betulaceae	1
Quaking aspen	Populus tremuloides	Salicaceae	4
Red maple	Acer rubrum	Sapindaceae	1
Red pine	Pinus resinosa	Pinaceae	3
Red spruce	Picea rubens	Pinaceae	3
Sugar maple	Acer saccharum	Sapindaceae; Aceraceae in older texts	1
Tamarack	Larix laricina	Pinaceae	3
White ash	Fraxinus americana	Oleaceae	5
White oak	Quercus alba	Fagaceae	6
White spruce	Picea glauca	Pinaceae	3
Yellow birch	Betula alleghaniensis	Betulaceae	1

TAXONOMIC UNIVERSE

Master Species List - Tier B

Regional mastery - important trees that become common within particular portions or forest systems of the course region. 44 taxa.

Preferred common name	Canonical scientific name	Family	Week
American mountain-ash	Sorbus americana	Rosaceae	4
American sycamore	Platanus occidentalis	Platanaceae	5
Atlantic white-cedar	Chamaecyparis thyoides	Cupressaceae	8
Balsam poplar	Populus balsamifera	Salicaceae	4
Bitternut hickory	Carya cordiformis	Juglandaceae	7
Black maple	Acer nigrum	Sapindaceae	2
Black oak	Quercus velutina	Fagaceae	6
Black walnut	Juglans nigra	Juglandaceae	7
Black willow	Salix nigra	Salicaceae	5
Blackgum	Nyssa sylvatica	Nyssaceae; sometimes Cornaceae	7
Boxelder	Acer negundo	Sapindaceae	5
Bur oak	Quercus macrocarpa	Fagaceae	6
Butternut	Juglans cinerea	Juglandaceae	7
Chestnut oak	Quercus montana	Fagaceae	6
Chokecherry	Prunus virginiana	Rosaceae	2
Common hackberry	Celtis occidentalis	Cannabaceae	6
Cucumbertree	Magnolia acuminata	Magnoliaceae	7
Eastern cottonwood	Populus deltoides	Salicaceae	5
Eastern redcedar	Juniperus virginiana	Cupressaceae	3
Flowering dogwood	Cornus florida	Cornaceae	7
Gray birch	Betula populifolia	Betulaceae	2
Heart-leaved paper birch	Betula cordifolia	Betulaceae	2
Jack pine	Pinus banksiana	Pinaceae	3
Mockernut hickory	Carya tomentosa	Juglandaceae	7
Mountain maple	Acer spicatum	Sapindaceae	2
Northern pin oak	Quercus ellipsoidalis	Fagaceae	6
Pagoda dogwood	Cornus alternifolia	Cornaceae	2
Pignut hickory	Carya glabra	Juglandaceae	7
Pin cherry	Prunus pensylvanica	Rosaceae	2
Pin oak	Quercus palustris	Fagaceae	6
Pitch pine	Pinus rigida	Pinaceae	8
River birch	Betula nigra	Betulaceae	5
Rock elm	Ulmus thomasii	Ulmaceae	5
Sassafras	Sassafras albidum	Lauraceae	7

Preferred common name	Canonical scientific name	Family	Week
Scarlet oak	Quercus coccinea	Fagaceae	6
Shagbark hickory	Carya ovata	Juglandaceae	7
Showy mountain-ash	Sorbus decora	Rosaceae	4
Silver maple	Acer saccharinum	Sapindaceae	5
Slippery elm	Ulmus rubra	Ulmaceae	5
Speckled alder	Alnus incana subsp. rugosa	Betulaceae	4
Striped maple	Acer pensylvanicum	Sapindaceae	2
Swamp white oak	Quercus bicolor	Fagaceae	6
Sweet birch	Betula lenta	Betulaceae	2
Tuliptree	Liriodendron tulipifera	Magnoliaceae	7

TAXONOMIC UNIVERSE

Master Species List - Tier C

Recognition and confusers - peripheral, planted, invasive, uncommon, or especially useful comparison taxa. 10 taxa.

Preferred common name	Canonical scientific name	Family	Week
American chestnut	Castanea dentata	Fagaceae	7
Black locust	Robinia pseudoacacia	Fabaceae	9
European larch	Larix decidua	Pinaceae	3
Norway maple	Acer platanoides	Sapindaceae	2
Norway spruce	Picea abies	Pinaceae	3
Scots pine	Pinus sylvestris	Pinaceae	3
Siberian elm	Ulmus pumila	Ulmaceae	5
Smooth serviceberry	Amelanchier laevis	Rosaceae	2
Tree-of-heaven	Ailanthus altissima	Simaroubaceae	9
White mulberry	Morus alba	Moraceae	9

FIELD DISCRIMINATION

Major confusion sets

- Sugar maple / black maple / Norway maple
- Red maple / silver maple; striped maple / mountain maple
- White ash / green ash / black ash
- Paper birch / heart-leaved paper birch / gray birch; yellow birch / sweet birch
- White spruce / black spruce / red spruce / Norway spruce; spruce / fir / hemlock
- Red pine / jack pine / Scots pine; tamarack / European larch
- Northern red oak / black oak / scarlet oak / northern pin oak / pin oak
- White oak / swamp white oak / bur oak / chestnut oak
- Quaking aspen / bigtooth aspen / balsam poplar / eastern cottonwood
- American elm / slippery elm / rock elm / Siberian elm
- Black cherry / pin cherry / chokecherry
- American hornbeam / eastern hophornbeam
- Black walnut / butternut; bitternut / shagbark / pignut / mockernut hickories
- Northern white-cedar / Atlantic white-cedar / eastern redcedar
- Native compound-leaved trees / black locust / tree-of-heaven

At every serious comparison, ask four things: What is it? What is the nearest plausible alternative? Which visible character separates them? What additional view would resolve uncertainty?

Begin the course

The full course is available free at cochetopa.co/course/. Move at your own pace, return as often as needed, and let unfamiliar specimens - not a calendar - determine when a species is actually learned.